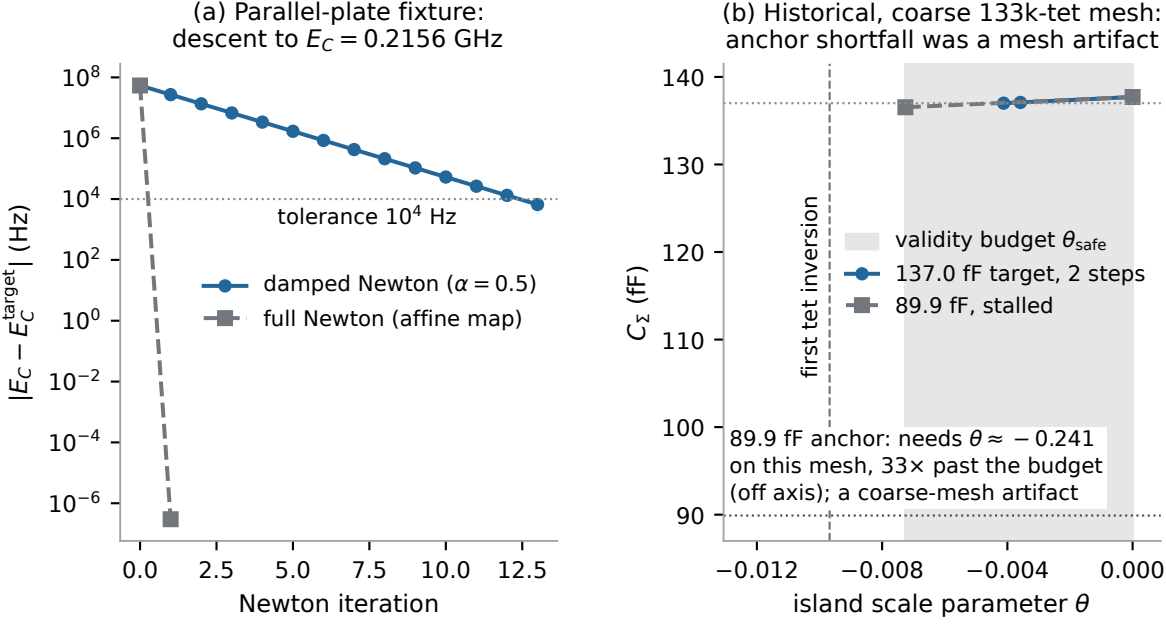
(a) Parallel-plate fixture:
descent to $E_C = 0.2156$ GHz
$|E_C - E_C^{\text{target}}|$ (Hz)
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
tolerance 10^4 Hz
damped Newton ($\alpha = 0.5$)
full Newton (affine map)
0.0
2.5
5.0
7.5
10.0
12.5
Newton iteration
(b) Historical, coarse 133k-tet mesh:
anchor shortfall was a mesh artifact
C_Σ (fF)
140
130
120
110
100
90
first tet inversion
validity budget θ_{safe}
137.0 fF target, 2 steps
89.9 fF, stalled
89.9 fF anchor: needs $\theta \approx -0.241$
on this mesh, 33× past the budget
(off axis); a coarse-mesh artifact
−0.012
−0.008
−0.004
0.000
island scale parameter θ